PENETRATION TESTING REPORT

Footprinting, Network Scanning and Vulnerability Assessment

1. PROJECT INFORMATION

Field	Details
Author	Osagie Osereme
Role	Cybersecurity Intern
Organization	Networkwalks
Program	Cybersecurity Internship
Batch	B084
Week	Week 02
Date	12 November 2026
Target	scanme.nmap.org (authorized test target)

2. EXECUTIVE SUMMARY

This report documents footprinting, network scanning and a basic vulnerability assessment carried out against an authorized target. Passive and active reconnaissance tools were used to collect domain, DNS, web technology and open-port information, which was then reviewed for security weaknesses. Summarize your most important finding here.

3. SCOPE AND AUTHORIZATION

- **In scope:** scanme.nmap.org
- **Out of scope:** any other domain or IP address
- **Authorization:** the target explicitly permits scanning for testing purposes

4. TOOLS USED

Tool	Purpose
whois	Domain registration details
nslookup	DNS record lookup
dnsrecon	DNS enumeration
whatweb	Web technology fingerprinting
wafw00f	Web application firewall detection
curl	HTTP header inspection

Tool	Purpose
nmap	Port and service scanning
Zenmap	GUI for Nmap, topology view

5. METHODOLOGY

1. Passive reconnaissance (footprinting)
2. Active reconnaissance (DNS, web fingerprinting)
3. Network scanning (ports, services, OS)
4. Vulnerability assessment
5. Reporting and recommendations

6. FOOTPRINTING

6.1 WHOIS Lookup

Command:

```
whois scanme.nmap.org
```

Findings: registrar, creation and expiry dates, name servers.

[Insert screenshot: whoisresult.png]

6.2 NSLookup

Command:

```
nslookup scanme.nmap.org
```

Findings: IP address(es) and record types.

[Insert screenshot: nslookup.png]

6.3 DNS Recon

Command:

```
dnsrecon -d nmap.org
```

Findings: A, MX, NS, TXT records and subdomains.

[Insert screenshot: dnsrecon.png]

7. WEB FINGERPRINTING

7.1 WhatWeb

Command:

```
whatweb scanme.nmap.org
```

Findings: server software, frameworks and versions.

[Insert screenshot: whatweb.png]

7.2 WAF Detection

Command:

```
wafw00f http://scanme.nmap.org
```

Findings: whether a WAF is detected.

[Insert screenshot: wafwof.png]

7.3 HTTP Headers (curl)

Command:

```
curl -I http://scanme.nmap.org
```

Findings: server header; security headers present or missing.

[Insert screenshot: curl.png]

8. NETWORK SCANNING

8.1 Basic Nmap Scan

Command:

```
nmap scanme.nmap.org
```

Findings: open ports and services.

[Insert screenshot: nmap.png]

8.2 Service and Version Detection

```
nmap -sV -O scanme.nmap.org
```

Port	State	Service	Version
22/tcp	open	ssh	(fill in)
80/tcp	open	http	(fill in)

8.3 Zenmap Results

[Insert screenshot: zenmaphostdetails.png]

[Insert screenshot: zenmapportsandhost.png]

[Insert screenshot: zenmapintensescan.png]

[Insert screenshot: zenmaptopology.png]

9. VULNERABILITY ASSESSMENT

#	Finding	Severity	Evidence	Recommendation
1	Outdated service version	Medium	Nmap -sV output	Update to latest version
2	Missing security headers	Low	curl -I output	Add HSTS, CSP, X-Frame-Options
3	Unnecessary open port	Low	Nmap scan	Close or firewall the port

Severity scale: Critical, High, Medium, Low, Informational.

10. RISK SUMMARY

Severity	Count
Critical	0
High	0

Severity	Count
Medium	1
Low	2

11. RECOMMENDATIONS

1. Patch and update exposed services.
2. Restrict unnecessary ports with a firewall.
3. Add missing HTTP security headers.
4. Limit information leaked in banners and WHOIS records.

12. CONCLUSION

Briefly summarize what was learned and the overall security posture of the target.

13. REFERENCES

• Nmap documentation: https://nmap.org/docs.html
• OWASP Testing Guide: https://owasp.org/www-project-web-security-testing-guide/

Disclaimer: This assessment was performed for educational purposes on systems I was authorized to test.